

CDS 101 – Final Project Report

Cntrl Alt Del

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Replace this template text with your own writing, keeping the headings and overall structure aligned with the **CDS 101 project rubric**.

1. Problem Definition

The problem we are investigating is if diversity has an effect on retention rates in American 4 year college, our data comes from the college.rds data set which has a plethora of information about American colleges, our study is based on American 4 year universities. I think the question is important because it can show students more information about their college and how that correlates to how others feel about it; which can help students make more informed decisions about their college choices.

Our goal is to find out whether or not higher diversity has a positive or negative correlation if any on retention rate.

Our objectives are to: Analyze 4 year universities Compare their diversity Compare their retention rates Analyze any correlation if there is any

2. Data Acquisition & Description

The dataset we are using is called college.rds

We obtained it through a download link from canvas. The variables we are using are UGDS_WHITE- Total share of enrollment of undergraduate degree-seeking students who are white UGDS_BLACK- Total share of enrollment of undergraduate degree-seeking students who are black UGDS_HISP- Total share of enrollment of undergraduate degree-seeking students who are Hispanic UGDS_ASIAN- Total share of enrollment of undergraduate degree-seeking students who are Asian UGDS_AIAN- Total share of enrollment of undergraduate degree-seeking students who are American Indian/Alaska Native UGDS_NHPI- Total share of enrollment of undergraduate degree-seeking students who are Native Hawaiian/Pacific Islander RET_FT4- The retention rate of 4 year universities in America The original dataset has 2228 columns, and 5 columns, our dataset has 6614 rows and 7 columns.

3. Data Cleaning & Preprocessing

Describe the steps you took to make the data usable, for example: The first thing we did to make the data more usable was narrowing down the data set and making the 'college_reduced' data set.

Next we renamed the variables that we were using to make them easier to understand.

We worked to create new groupings like pct_nonwhite, which made it easier to show if a school has a higher or lower diversity

Another column was added to name the regions of the schools, and another to sort the types of diversity as higher or lower.

Another problem that we had to work through was working through NA's in the data analysis section, which is what 'is.na' (Filters out rows with NAs) and 'na.rm = TRUE' (ignores NAs when calculating) are for.

```
college_reduced <- college %>%
  select(
    UNITID,
    RET_FT4,
    UGDS_WHITE,
    UGDS_BLACK,
    UGDS_HISP,
    UGDS_ASIAN,
    UGDS_AIAN,
    UGDS_NHPI,
    REGION
  )
```

```
college_reduced <- college_reduced %>%
  rename(
    retention_rate = RET_FT4,
    pct_white = UGDS_WHITE,
    pct_black = UGDS_BLACK,
    pct_hispanic = UGDS_HISP,
    pct_asian = UGDS_ASIAN,
    pct_aian = UGDS_AIAN,
    pct_nhpi = UGDS_NHPI,
    region = REGION
  )
```

```
college_reduced <- college_reduced %>%
  mutate(
    pct_nonwhite = pct_black + pct_hispanic + pct_asian + pct_aian + pct_nhpi
  )
```

```
college_reduced <- college_reduced %>%
  mutate(
    region_name = recode(region,
      `0` = "U.S. Service Schools",
      `1` = "New England",
      `2` = "Mid East",
      `3` = "Great Lakes",
      `4` = "Plains",
      `5` = "Southeast",
      `6` = "Southwest",
      `7` = "Rocky Mountains",
      `8` = "Far West",
      .default = NA_character_
    )
  )
```

```
college_reduced <- college_reduced %>%
  mutate(
    diversity_group = if_else(
      pct_nonwhite > median(pct_nonwhite, na.rm = TRUE),
      "Higher Diversity",
      "Lower Diversity"
    )
  )
```

4. Exploratory Data Analysis (EDA)

The exploratory data analysis focuses on the relationship between diversity and retention rate. The main variables examined are retention_rate, pct_nonwhite, and diversity_group. These graphs and summary statistics help show whether higher diversity schools appear to have different retention patterns than lower diversity schools.

The summary table compares retention rates between higher diversity and lower diversity universities.

```
summary_stats <- college_reduced %>%
  group_by(diversity_group) %>%
  summarize(
    count = n(),
    mean = mean(retention_rate, na.rm = TRUE),
    median = median(retention_rate, na.rm = TRUE),
    sd = sd(retention_rate, na.rm = TRUE),
    iqr = IQR(retention_rate, na.rm = TRUE),
    min = min(retention_rate, na.rm = TRUE),
    max = max(retention_rate, na.rm = TRUE)
  )
summary_stats
```

```
## # A tibble: 3 x 8
##   diversity_group count    mean median    sd    iqr  min  max
##   <chr>          <int>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 Higher Diversity 3155   0.665  0.706  0.219  0.257    0    1
## 2 Lower Diversity 3156   0.745  0.770  0.158  0.175    0    1
## 3 <NA>           747  NaN    NA    NA    NA    Inf -Inf
```

The summary statistics show noticeable differences in retention rates between the two diversity groups. Lower diversity schools had a higher average retention rate of about 0.745 compared to about 0.665 for higher diversity schools. The median retention rate was also higher for lower diversity schools. Higher diversity schools also had greater variability, shown by the larger standard deviation and interquartile range.

The first graph compares the distribution of retention rates between higher diversity and lower diversity schools.

```
ggplot(college_reduced, aes(x = retention_rate, fill = diversity_group)) +
  geom_histogram(
    aes(y = after_stat(density)),
    binwidth = 0.05,
    position = "identity",
    alpha = 0.5
  )
```

```
) +
labs(
  title = "Distribution of Retention Rates by Diversity Group",
  x = "Retention Rate",
  y = "Density",
  fill = "Diversity Group"
)
```

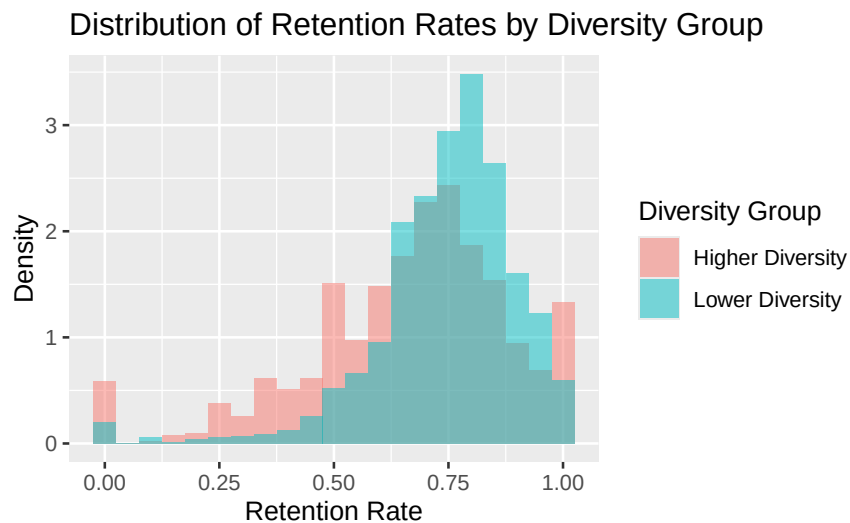


Figure 1: Retention rate distribution based on diversity level

This graph shows that both groups have many schools with retention rates between about 0.6 and 0.9. However, the lower diversity group appears more concentrated around higher retention values, while the higher diversity group has a wider spread and includes more schools with lower retention rates.

5. Visualization Quality and Storytelling

The histogram is an appropriate graph because it compares the distribution of retention rates between two sources, we used 'fill = diversity_group' to give different colors to differentiate between higher and lower diversity rates. Setting alpha to 0.5 makes the overlapping bars transparent so you can see both sets at the same time.

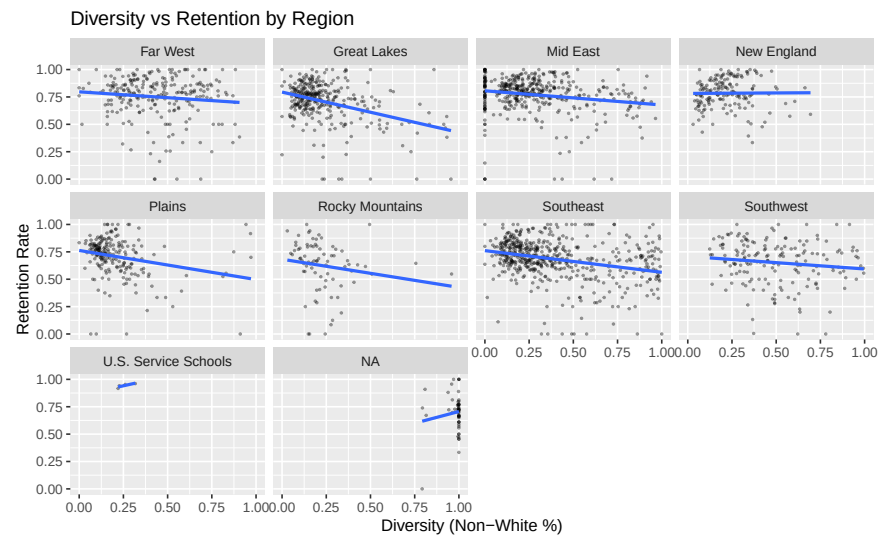
The faceted scatterplot graph is the correct graph for looking at the amount of diversity across different regions. We faceted by region name which makes each panel reasonable, and to make the data easier to read we used 'geom_smooth(method = "lm")' to add a trend line to each graph, which shows coorelation.

```
ggplot(college_reduced, aes(x = pct_nonwhite, y = retention_rate)) +
  geom_point(alpha = 0.4, size = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  facet_wrap(~region_name) +
  labs(
    title = "Diversity vs Retention by Region",
    x = "Diversity (Non-White %)",
    y = "Retention Rate"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 4898 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 4898 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



This code chunk is here to find the mean retention rates of schools with lower and higher diversity, and it shows that lower diversity schools have a higher mean retention rate.

```
college_reduced %>%
  group_by(diversity_group) %>%
  summarize(mean_retention = mean(retention_rate, na.rm = TRUE))
```

```
## # A tibble: 3 x 2
##   diversity_group mean_retention
##   <chr>           <dbl>
## 1 Higher Diversity    0.665
## 2 Lower Diversity    0.745
## 3 <NA>              NaN
```

This code chunk is here to fix errors with NA's in the null distribution code chunk, which was causing it to see 3 columns instead of two, and removes the third column that was made up of NA's, the NA's could have been schools with a very low nonwhite population, or schools that don't keep track of their student's demographics.

```
college_reduced <- college_reduced %>%
  filter(!is.na(diversity_group))
```

This code chunk exists to create the null distribution, which creates 10,000 differences in means that assume that diversity has no effect on retention rate.

```

null_distribution <- college_reduced %>%
  specify(retention_rate ~ diversity_group) %>%
  hypothesize(null = "independence") %>%
  generate(10000, type = "permute") %>%
  calculate(stat = "diff in means", order = c("Higher Diversity", "Lower Diversity"))

```

```
## Warning: Removed 4151 rows containing missing values.
```

this is just here to show the null distribution simulations

```
null_distribution
```

```

## Response: retention_rate (numeric)
## Explanatory: diversity_group (factor)
## Null...
## # A tibble: 10,000 x 2
##   replicate    stat
##   <int>    <dbl>
## 1         1 -0.0143
## 2         2  0.0111
## 3         3  0.0132
## 4         4  0.0111
## 5         5  0.0209
## 6         6  0.00605
## 7         7  0.00449
## 8         8  0.00197
## 9         9 -0.00963
## 10        10 -0.00600
## # i 9,990 more rows

```

This code chunk is here to calculate the actual difference in means (-0.080404), and then store it as 'observed_stat', so we can use it for the p-value code and correlating graph

```

observed_stat <- 0.6650291 - 0.7454331
observed_stat

```

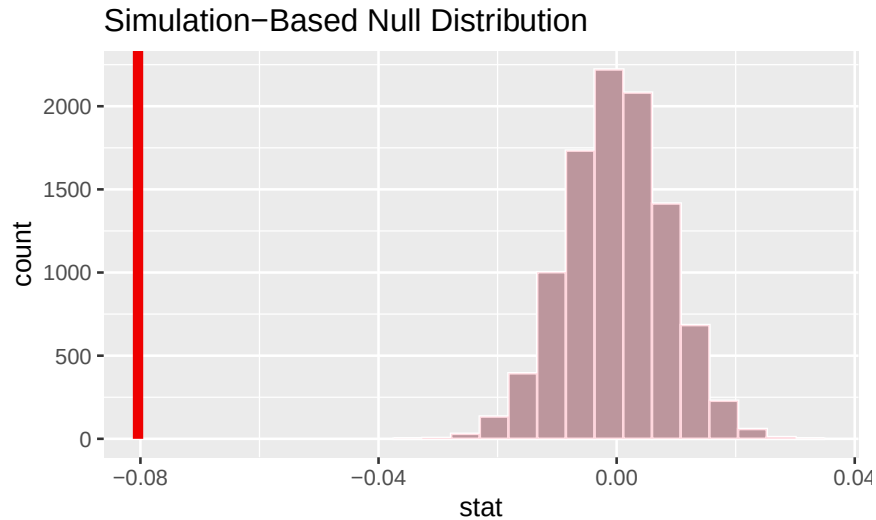
```
## [1] -0.080404
```

The null distribution plot is a histogram type of graph, and shows what the differences in mean would look like if it was due to chance alone. This graph was to visualize the null distribution, and helps clarify why we got a p-value of one.

```

visualize(null_distribution) +
  shade_p_value(obs_stat = observed_stat, direction = "greater")

```



Since we got a P-value of one we fail to reject the null hypothesis of diversity having no effect on college retention rate. # 6. Modeling Approach

Explain how you framed the problem and which models you chose:

For this project, we used statistical inference to investigate whether diversity has an effect on retention rates in American 4-year colleges.

The main analysis focused on a difference in means hypothesis test comparing retention rates between higher diversity and lower diversity schools. Schools were grouped based on whether their `pct_nonwhite` value was above or below the median.

We created a null distribution using permutation methods with the `infer` package to test whether the observed difference in retention rates could have occurred by chance.

In addition to the hypothesis testing, simple linear regression trend lines were added to the scatterplots using `geom_smooth(method = "lm")` to help visualize the relationship between diversity and retention rate across regions

7. Model Implementation & Evaluation

This corresponds to the **Model Implementation & Evaluation** rubric criterion.

The statistical inference analysis used `retention_rate` as the response variable and `diversity_group` as the explanatory variable. Schools were divided into “Higher Diversity” and “Lower Diversity” groups based on whether their `pct_nonwhite` value was above or below the median.

Before running the analysis, rows with missing values in `diversity_group` were removed so the hypothesis test and null distribution could run correctly..

```
# Example structure:
# set.seed(123)
# train_index <- sample(seq_len(nrow(data_clean)), size = 0.8 * nrow(data_clean))
# train <- data_clean[train_index, ]
# test <- data_clean[-train_index, ]
#
# model <- glm(target ~ ., data = train, family = binomial)
```

```
# preds <- predict(model, newdata = test, type = "response")
# # compute metrics...
```

8. Conclusions & Recommendations

Based on the testing of our hypothesis we found no evidence that suggests that diversity has a positive effect on retention rate. Our observed statistic which was -0.080404 showed that schools with higher diversity actually had lower retention rates. This pattern is evident when looking at the facted scatterplots you can see the negative relationship between diversity and retention rates.

One of our biggest limitations was the amount of NA's(schools without reports on diversity) in the dataset which could have skewed the dataset, another limitation is that retention rate has more influences than just diversity rate, which means those schools could have external factors effecting retention rate.

Another limitation that could apply for a study like this is the timeframe of the dataset, instead of a dataset that isn't just a snapshot, but showing how diversity and retention rates change could provide more data.

In the future this could be done by looking at each racial group separately instead of combining them, and having more control for other variables of external factors, and statistics over a timeline could give more accurate data.

9. Code Quality & Reproducibility

To reproduce this project, someone would need the college.rds dataset and this RMarkdown file. The dataset should be saved in the same folder as the RMarkdown file so the read_rds("college.rds") function works correctly. The required R packages for this project are tidyverse, infer, modelr, and broom. These packages are loaded in the setup chunk at the beginning of the project. The project can be reproduced by running the RMarkdown file from top to bottom in order. This includes the setup section, data cleaning and preprocessing, exploratory data analysis, visualization, statistical inference testing, and conclusion sections. A seed was also set using set.seed(98261936) so the null distribution simulations and statistical results can be reproduced consistently.

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
```

```
##
## other attached packages:
## [1] broom_1.0.12      modelr_0.1.11      infer_1.1.0        lubridate_1.9.4
## [5] forcats_1.0.1      stringr_1.6.0      dplyr_1.1.4         purrr_1.2.1
## [9] readr_2.1.6        tidyr_1.3.2        tibble_3.3.1        ggplot2_4.0.1
## [13] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] utf8_1.2.6          generics_0.1.4      lattice_0.22-7      stringi_1.8.7
## [5] hms_1.1.4           digest_0.6.39       magrittr_2.0.4      evaluate_1.0.5
## [9] grid_4.5.2          timechange_0.4.0    RColorBrewer_1.1-3 fastmap_1.2.0
## [13] Matrix_1.7-4        backports_1.5.0     mgcv_1.9-3          scales_1.4.0
## [17] cli_3.6.5           rlang_1.1.7         crayon_1.5.3        splines_4.5.2
## [21] withr_3.0.2         yaml_2.3.12         otel_0.2.0          tools_4.5.2
## [25] tzdb_0.5.0          vctrs_0.7.1         R6_2.6.1            lifecycle_1.0.5
## [29] pkgconfig_2.0.3     pillar_1.11.1       gtable_0.3.6        glue_1.8.0
## [33] xfun_0.56           tidyselect_1.2.1    rstudioapi_0.18.0   knitr_1.51
## [37] farver_2.1.2        htmltools_0.5.9     nlme_3.1-168        rmarkdown_2.30
## [41] labeling_0.4.3      compiler_4.5.2      S7_0.2.1
```

10. References

List any references you used, such as:

emehex. (2015, March 4). Removing NA observations with `dplyr::filter()`. Stack Overflow. <https://stackoverflow.com/questions/28857653/removing-na-observations-with-dplyrfilter>

Ismay, C. (2025, September 16). Chapter 9 Hypothesis Testing | Statistical Inference via Data Science. Moderndive.com. <https://moderndive.com/9-hypothesis-testing.html>